

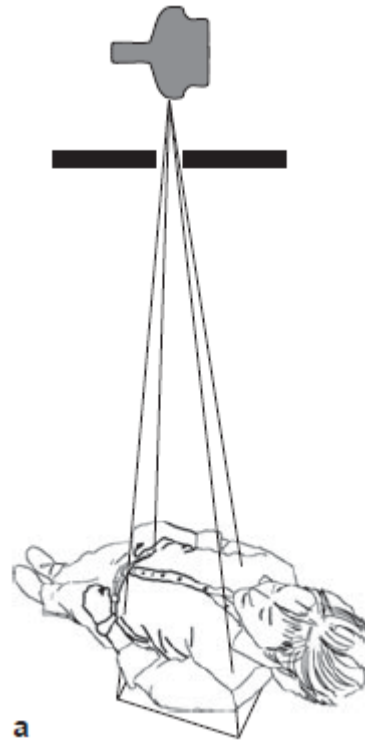
X-Ray based planar radiography
and computed tomography (CT)

X-ray Planar Radiography

- 2D
- Analog and Digital recording available
- Portable

Key aspects:

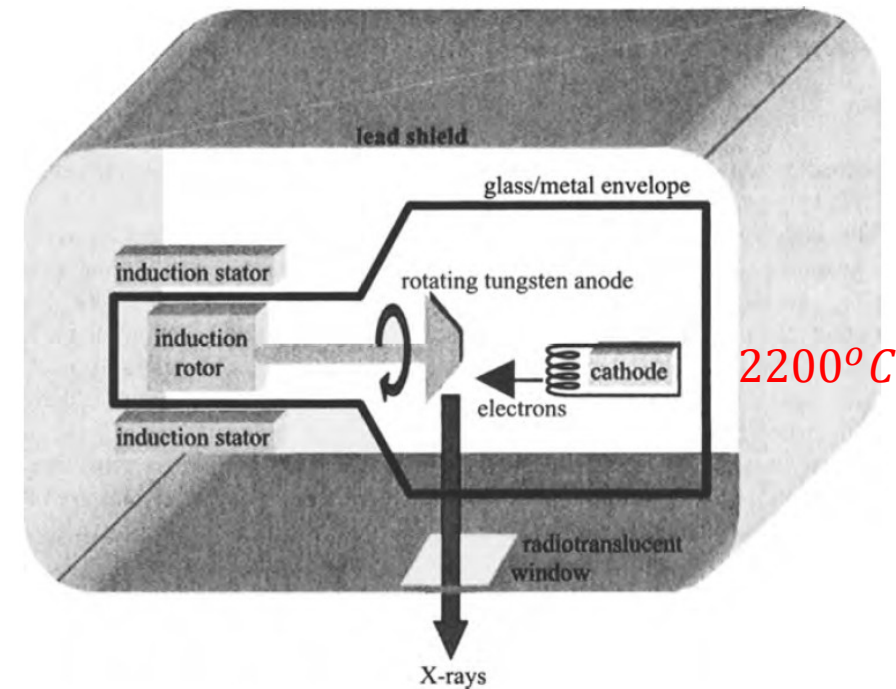
- X-ray generation (the X-ray Tube)
- X-ray interactions with the body
- X-ray detectors
- Factors affecting Resolution
- Signal to noise ratio
- Contrast



X-ray Tube

Peak potential difference between the Cathode and the Anode– is usually between 25 kV to 140 kV

- Thermionic emission of electrons from the Cathode forms a tube current that hits the anode which is at a positive potential (KVp)
- Both Cathode and Anode are usually made of Tungsten
 - High thermal conductivity
 - High melting point
 - Low Vapor Pressure
- Bevel angle is to achieve a small focal spot size



F is width of electron beam

$$f = F \sin \theta,$$

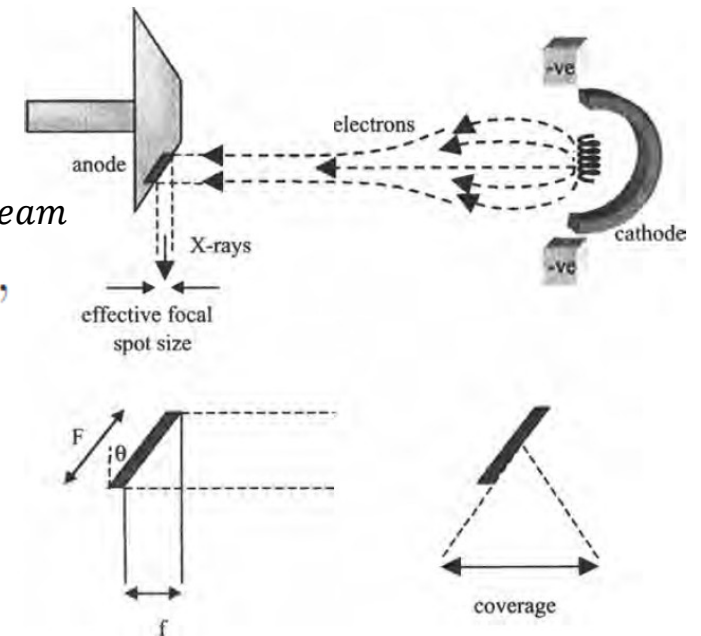


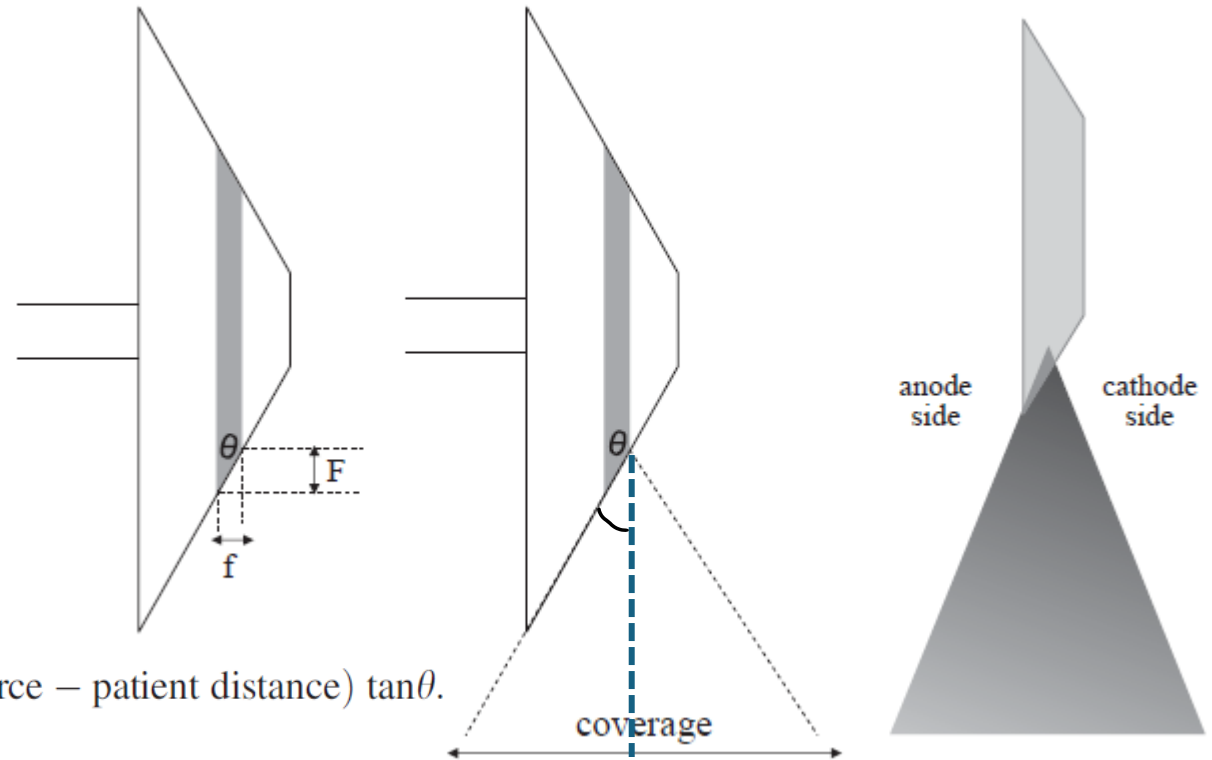
Figure 1-5: The Rotating Anode



The focal track (i.e. the location of the electron interactions) can be seen as the darker gray stripe ringing the edge of the anode disk.

Molybdenum is also used for Anode

$$\text{Coverage} = 2(\text{source} - \text{patient distance}) \tan \theta.$$

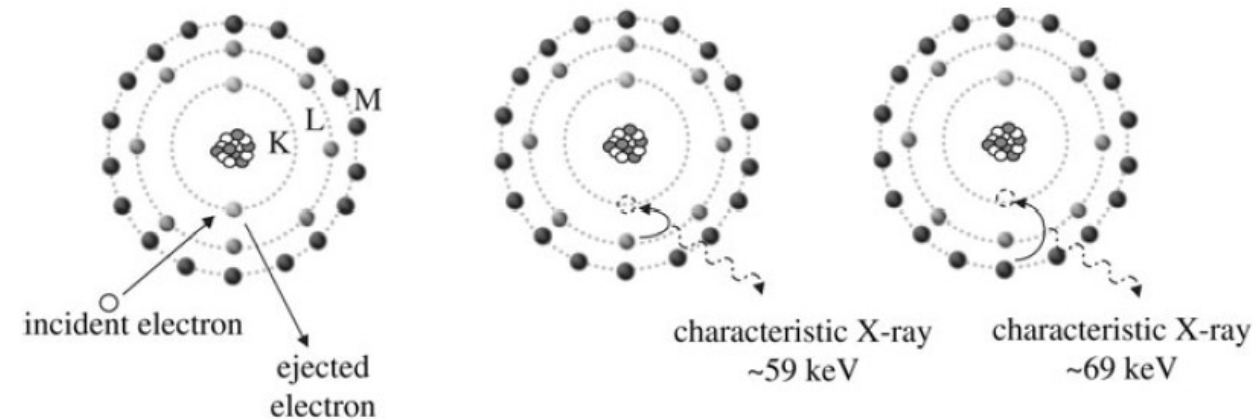
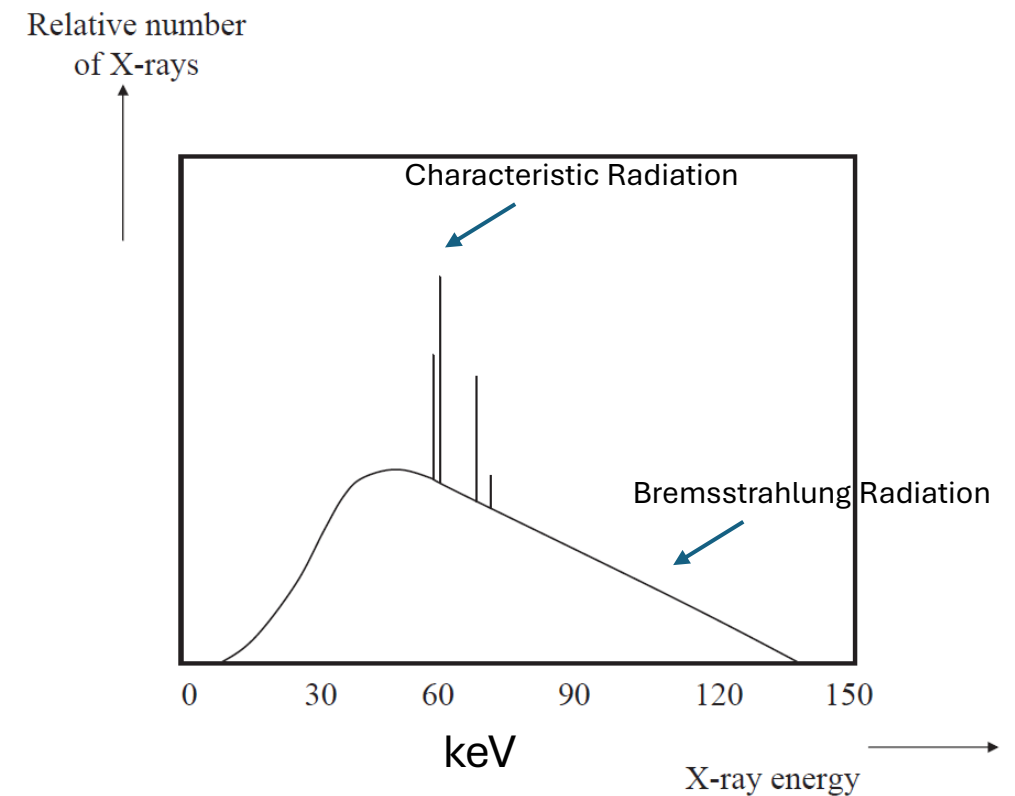
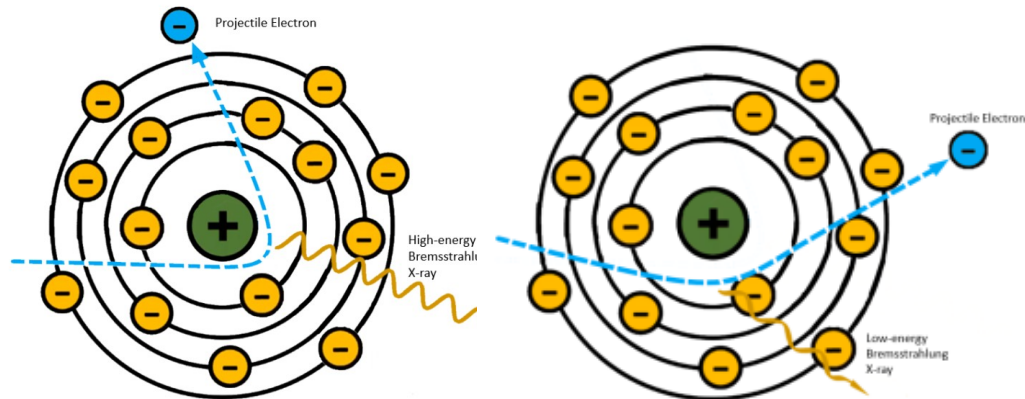


Heel effect: X-ray intensity on the cathode side is higher than that on anode side as the rays pass longer distance within the anode material on the anode side.

- **User controlled parameters** - kVp, tube current, and exposure time, based on the application.
- Tube-current range – 50mA to 400 mA for X-rays and 1000mA for CT.
- Heat-dissipation in the Anode is the limiting factor for achieving high tube-currents.

Mechanisms leading to X-ray radiation

- Bremsstrahlung Radiation (Braking Radiation)
 - Deceleration of the electrons within the anode material due to the nuclear-field
- Characteristic radiation (specific frequencies, characteristic of the anode material):
 - Radiation resulting from electron transitions from a higher energy shell (L or M) to a lower one. The transition occurs because holes are created when an incident electron (from the tube current) knocks an electron off from a lower shell an atom in the anode.



X-Ray interactions within the tissue

- Photoelectric attenuation
 - Energy of the incident X-ray is lost in entirety in the tissue, as it gets used up for knocking off electrons usually from low energy shells of the atoms in the tissues (Carbon, Oxygen, Calcium etc.)
- The probability (P_{pe}) of a photoelectric interaction occurring for an x-ray within the tissue depends on its
 - Energy (E)
 - Effective Atomic number of the tissue (Z_{eff})
 - Density of the tissue (ρ)

$$P_{pe} \propto \rho \frac{Z_{eff}^3}{E^3}$$

In generally, about 90% or more of the incident X-rays are absorbed in the body

EFFECTIVE ATOMIC NUMBER AND ENERGY ABSORPTION IN TISSUES*		
By F. W. SPIERS, Ph.D. The General Infirmary, Leeds		
1946		

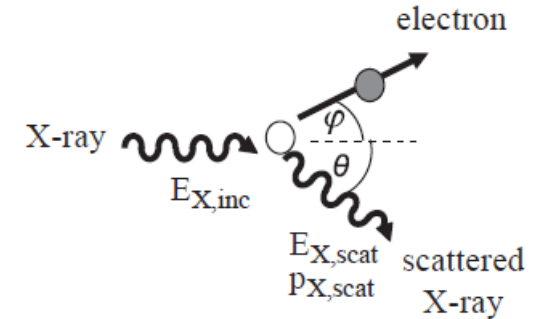
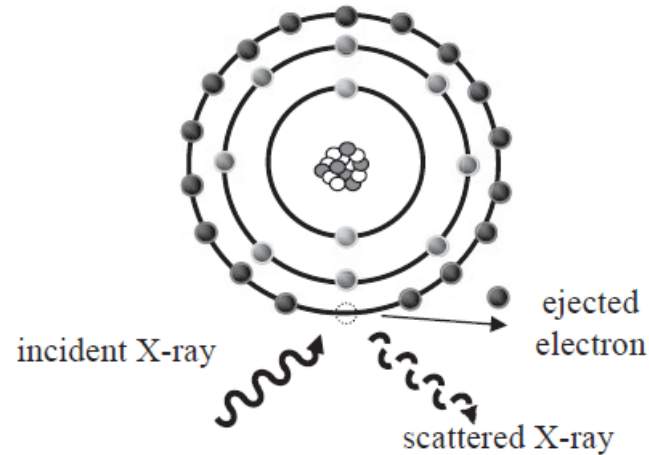
(1) Tissues	(6) $\bar{Z}$	(7) ρ gm./c.c. (
Muscle I (biceps)	7.42	1.00
Muscle II (biceps)	7.46	1.00
Muscle III* (gastrocnemius)	7.2	0.98
Skin	7.31	1.00
Subcutaneous fat	5.92	0.91
Animal fat* ..	5.6	0.90
Bone I (femur) · (as powder)	13.8	1.85
Bone II (femur) (cut section)	14.0	1.87
Bone III (animal) (as powder)	13.9	1.95

X-Ray interactions within the tissue

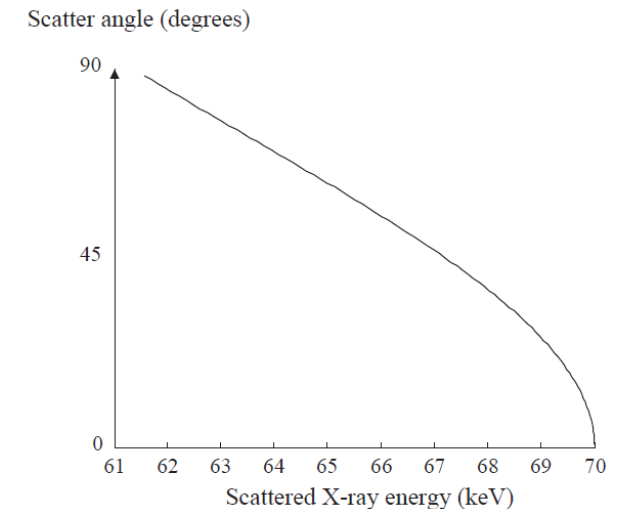
- Compton Scattering
 - Incident X-ray knocks off the electrons in outermost/valence shells and in the process loses some energy and deviates (gets scattered) from its path by an angle θ .
 - $\Delta\lambda$ between the incident ray and the scattered ray is given by

$$\Delta\lambda = \frac{h}{mc} (1 - \cos \theta)$$

$$\Delta E = \frac{E_{X,inc}}{1 + \frac{E_{X,inc}}{mc^2} \cdot (1 - \cos \theta)}$$



Electron density of the tissue is the other major factor that affects Compton Scattering



X-Ray interactions within the tissue

- Intensity of the X-rays is defined as the ‘power incident per unit area’.
- Power, in turn, depends on the total # of X-rays produced and the energy of the X-rays.
- The # of X-rays produced is proportional to the tube current.
- Energy of the X-rays produced is proportional to the square of the accelerating potential kVp.

$$I \propto (kVp)^2 \cdot (mA)$$

Radiation dose to the patient is proportional to the intensity of the X-rays and exposure time

X-Ray interactions within the tissue

- Attenuation of the X-rays as a function of distance, follows an exponential relationship

$$N = N_0 e^{-\mu(E) \cdot x}$$

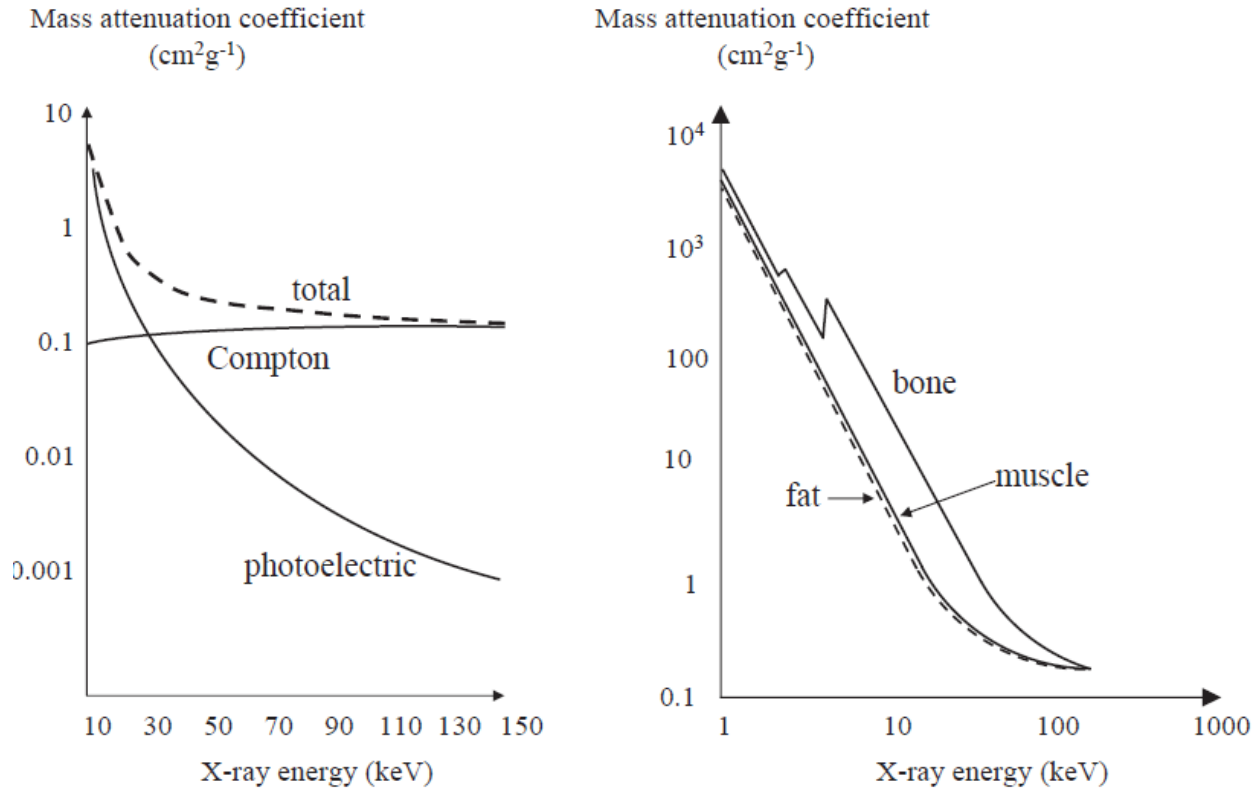
N_0 is the number of incident X-rays, $\mu(E)$ is the linear attenuation factor, which is dependent on the energy of the incident X-ray. μ is measured in (attenuation per cm^{-1} ($\mu = \frac{\ln(\frac{N_0}{N})}{x}$))

μ depends on the incident energy and the tissue type

$$\mu = \mu_{photoelectric} + \mu_{Compton}$$

μ/ρ – mass attenuation coefficient; units of cm^2/g

X-Ray interactions within the tissue



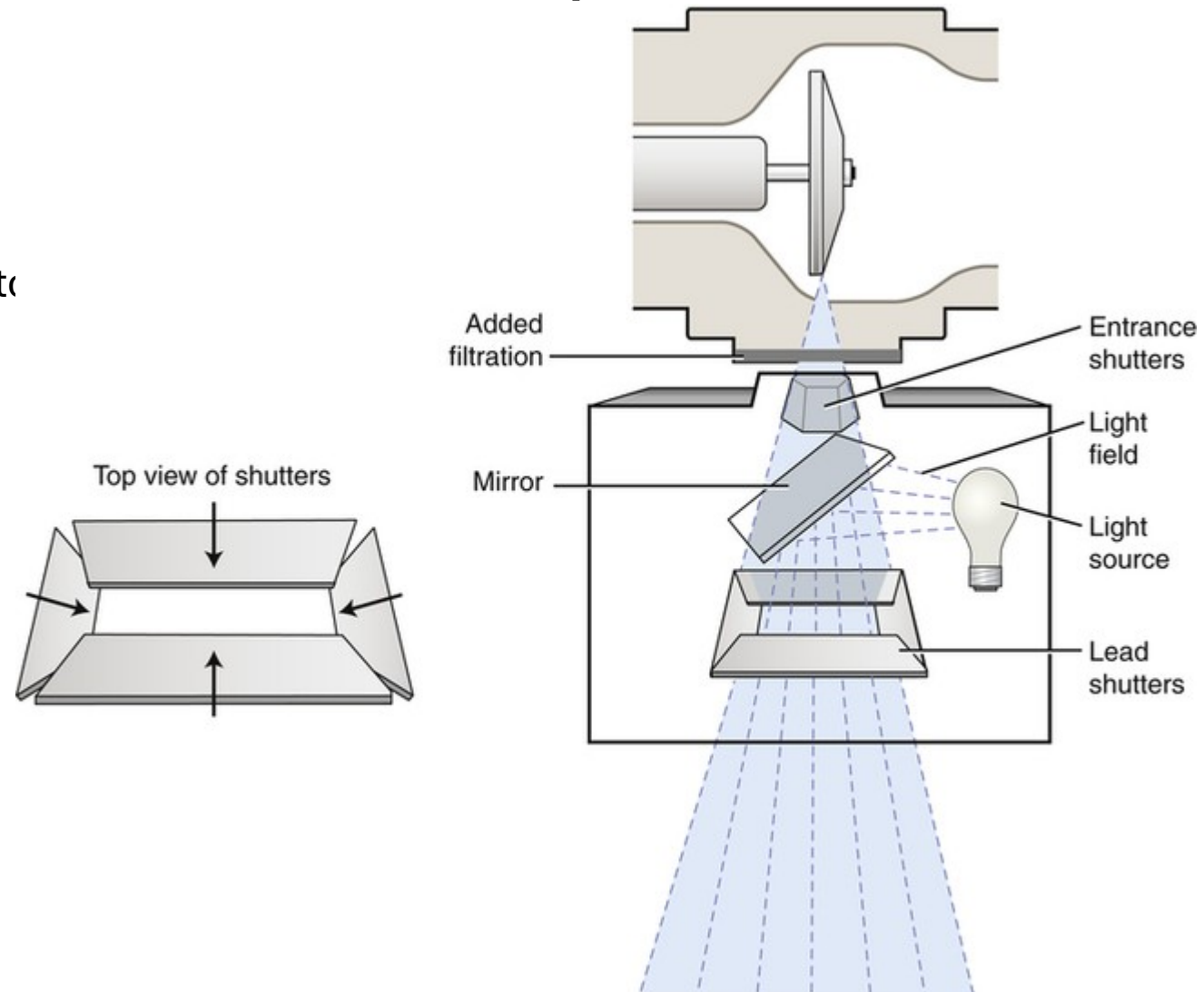
- At lower energies, absorption is high ($\frac{\mu}{g}$ is high)
- As energy of the incident X-ray increases, μ/g decreases
- The probability of photoelectric attenuation increases as the energy approaches the binding energy of the k-shell, there is an abrupt jump in the value of μ/g (i.e., increased absorption), called the *K-edge*.
- Another way of characterising X-ray absorption in a tissue is: Half Value Layer (HVL), - the tissue thickness at which the intensity of the incident X-ray reduces by $\frac{1}{2}$.

- $HVL = \ln(2) / \mu$

Example...@100 keV HVL of muscle is 3.9 cms

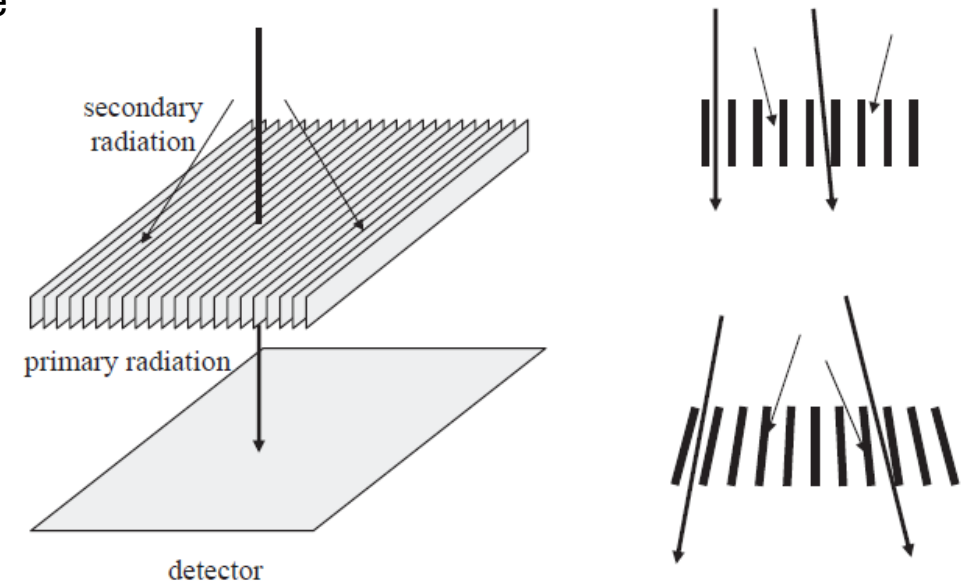
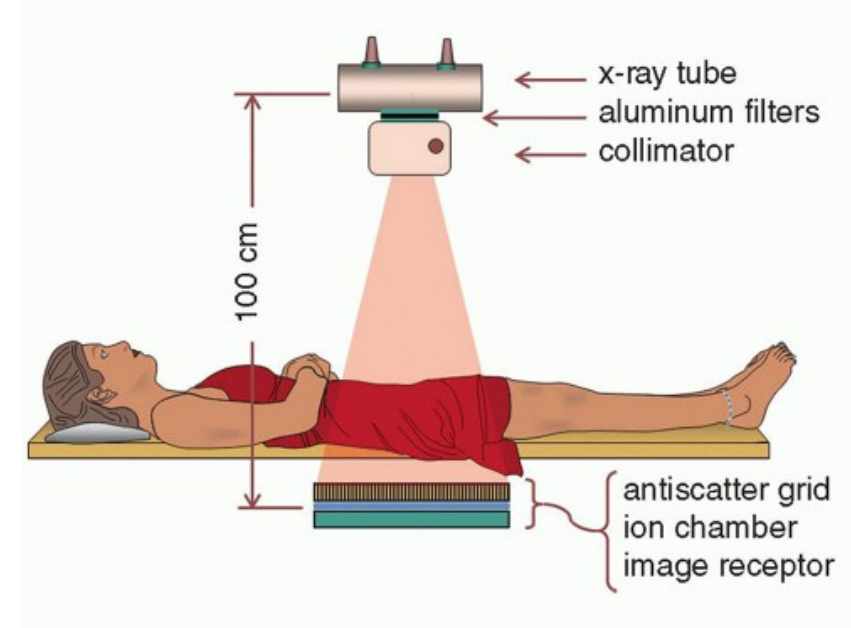
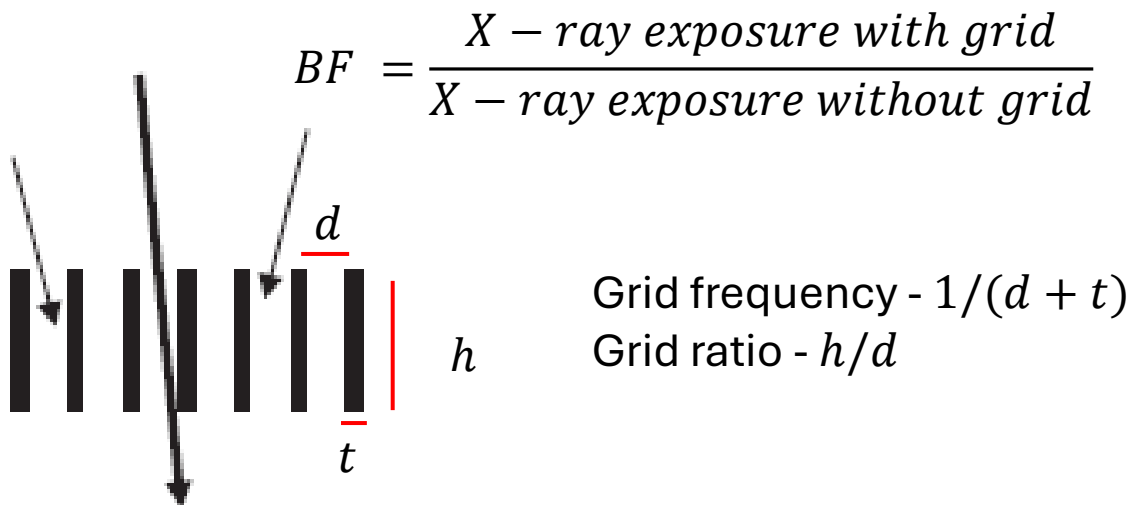
X-Ray Collimation (beam restrictor)

- Coverage is determined by the bevel angle.
- However, further restriction of the X-ray beam to the anatomy of interest (field of view), called collimation, is performed using lead-window
- Collimators or beam restrictors are placed between the beam source and the patient
- This reduces patient dose
- Also reduces scatter



X-Ray Anti-scatter grid

- Placed between the patient and the X-ray detector, it restricts the scatter radiation from reaching the X-ray detector/film.
- Reduces SNR (which is proportional to the # of x-rays reaching the detector)
- To compensate for this, Intensity is usually increased to obtain good SNR image.
- The factor by which patient dose gets increased, due to the use of an anti-scatter grid is called the **Bucky Factor**. (typical BF 4 to 10)





Without grid

*With grid and no increase in
patient exposure*

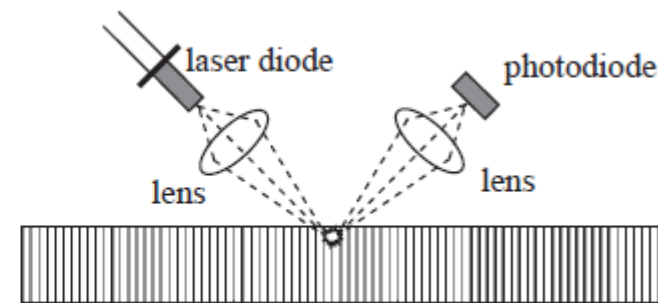
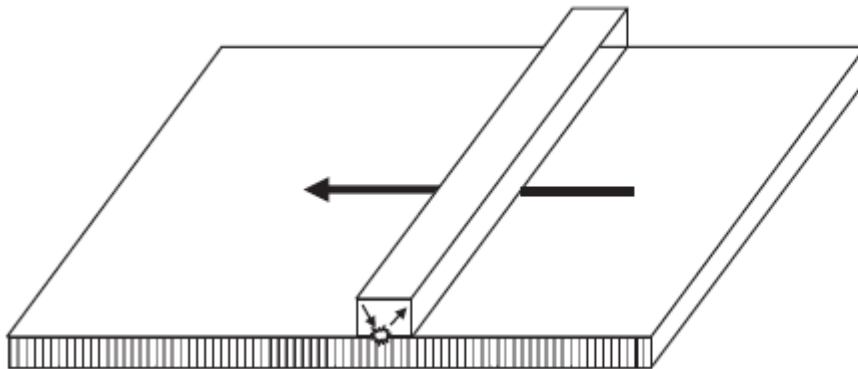


X-Ray Detectors

- Computed Radiography
- Digital Radiography

Computed Radiography detector

- Contains a **detector plate** which release electrons that are still trapped within the material.
- When the detector layer is exposed to optical laser, it causes most of these electrons to return to ground state, which in the process leads to emission of visible light. This is then detected using an array of **photo-diodes** that convert this to a voltage which is then amplified and digitized.
- Sampling can happen every $50\ \mu\text{m}$
- Dynamic range is large – spanning 4 orders of magnitude



X-Ray Detectors

- Computed Radiography
- Digital Radiography

Digital Radiography detectors

- Two types – Indirect and Direct
- Indirect: Uses scintillators (CsI:TI – Cesium Iodide doped with Thallium), which are materials that emit visible light when exposed to X-rays. They are used in combination with a 2D flat panel detectors, consisting of an array of photodiodes, a capacitor, and Thin Film Transistors.
- Direct: Uses detector materials (amorphous Selenium) that converts absorbed X-rays to electrical signal.

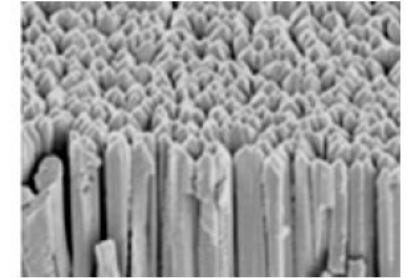
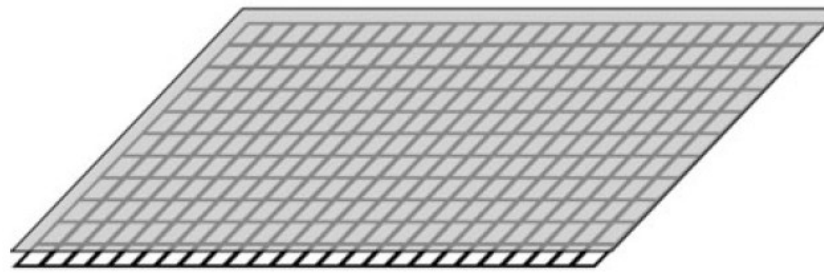
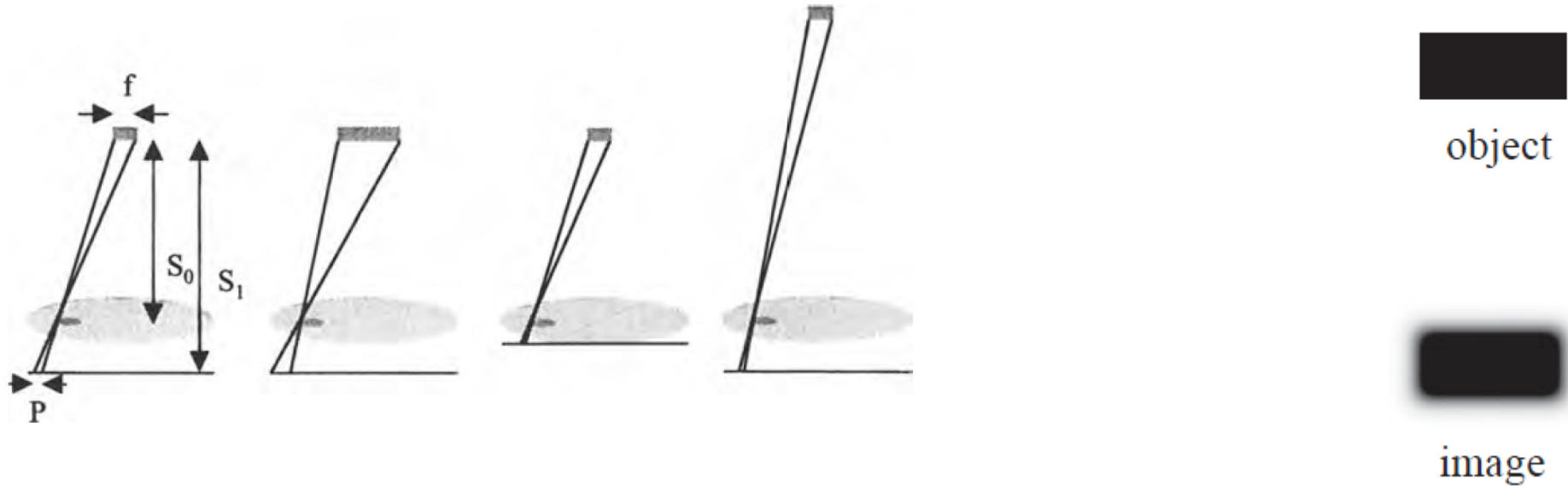


Figure 2.16

(left) A thin CsI:TI layer (shaded) placed on top of a SiH active matrix array. (right) An electron micrograph showing the needle-like structure of the crystals of CsI:TI.

Some additional factors affecting X-ray Image quality

Effect of focal spot size



- Noise standard deviation, σ , in the detector is proportional to $\sqrt{N}$, N being the incident # of X-rays